MediHaven: Integrated Multimodal Clinical Intelligence and Sovereign Medical Vault Platform

Vidhi [Surname] • Nirupam [Surname] • Param [Surname] | Guide: [Guide name and designation] | Group [Group No.]

Department of Computer Engineering, Sardar Patel Institute of Technology (Autonomous, University of Mumbai) • T.E. Mini Project (Semester V) • AY 2026–27

Educational prototype not clinically validated not for patient care

Problem

Deterioration is often recognised late; patients rarely control who sees their records; and published clinical ML results are hard to reproduce and can fail on new data (an externally validated sepsis model reached AUROC 0.63).

Objectives

- Explainable 4-model triage on synthetic inpatients
- Patient vault with scoped, expiring, single-use QR passes
- Honest MIMIC-IV Demo mortality pipeline (no leakage, CIs)
- Fail-closed, loopback-only research serving
- Secure, tested, measured, deployable

Technology Stack

Python 3.12 • Flask 3.1 • Waitress • SQLite (WAL) • scikit-learn 1.9 • pandas • DuckDB • PyJWT (HS256) • OpenCV • Chart.js 4.4 • pytest • Docker recipe • optional Google Gemini (de-identified prompts)

Data

Synthetic demo: 100 generated inpatients, 4,368 vitals rows, 30 engineered features.
Research: MIMIC-IV Clinical Database Demo 2.2 (open access). 140 ICU stays → **107 eligible**; label = in-hospital death after ICU admission + 24 h; **11 deaths** (4 with a death time after discharge, kept and flagged), 96 survivors. Data stay on the researcher's machine.

System Architecture

Browser portals: landing, login, physician triage, patient vault, QR scanner, admin, research (opt-in) — Jinja2, JavaScript, Chart.js

↓ HTTP + JSON

Flask application: session RBAC, CSP and security headers, rate limits, research-mode boundary; REST blueprints auth, patients, predictions, alerts, vault, research

Services: 4-model ensemble • vault and HMAC-signed QR passes • de-identified narrative synthesiser • research service (verified artifacts, out-of-fold stay view)

Data: SQLite (WAL, 8 tables) • joblib models • research artifacts outside Git/OneDrive • optional Gemini API (synthetic mode only)

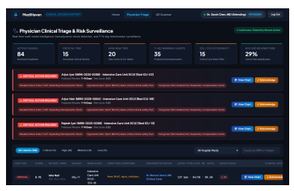
Main Workflows

Triage: roster → patient chart → consensus tier, each model's vote, decision-tree rule path, similar cases, 72-h vitals.

Vault: patient shares a pass (scope, lifetime ≤ 30 days, uses) → clinician scans → signature, expiry, revocation, *atomic* use quota → only consented records, access logged under the signed-in clinician.

Research (opt-in): build dataset → grouped CV → hash-verified run → analytics, what-if and stay-level triage with out-of-fold scores.

Screens (synthetic data)



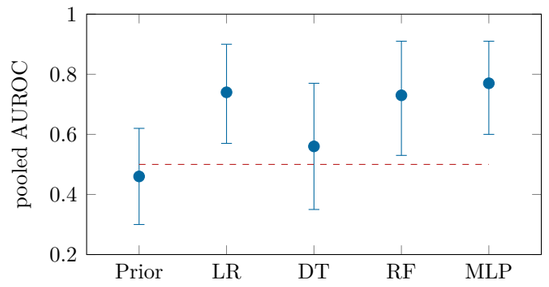


Models

Synthetic ensemble (4 tiers): weighted soft vote 0.30 decision tree +0.30 MLP +0.25 KNN +0.15 K-Means; tier = arg max, score = $\sum_t p_t s_t$.

Research baselines (74 inputs from the first 24 ICU hours): prior-only reference, L2 logistic regression, m-estimate-smoothed decision tree, random forest, small MLP; median imputation inside each fold.

Research Results: 5-fold grouped CV, 95% CI



107 stays, 11 deaths. Intervals overlap heavily; without the 4 anomalies (7 deaths, retrained) LR and MLP fall to about 0.40. **Pipeline validation only, no clinical performance claim.** Synthetic hold-out accuracy 1.00 reflects the data generator, not clinical skill.

Research Mode Safeguards

Explicit opt-in (`DATA_MODE=mimic`) • loopback only • strong secrets enforced • SHA-256-verified artifacts • never calls Gemini • stays numbered, no MIMIC identifiers or dates shown • research-only banners.

Testing

354 automated tests: 354 passed, 0 failed (pytest, 26 Oct 2026): unit, API integration, security, invalid-input, concurrency (4 simultaneous scans of a single-use pass → exactly 1 grant) and fail-closed research tests. Tests use temporary paths and never touch real data or secrets.

Measured Performance (Waitress, 8 threads)

Prediction latency, 1 client (p50)	71.9 ms
Prediction throughput (saturated)	≈15 req/s
Vitals reads, 16 clients	268.8 req/s
Errors in 3,200 requests	0
Server memory under load	≤ 206 MB

Bottleneck: CPU-bound feature engineering + 4 models per prediction.

Limitations and Intended Use

Synthetic data cannot show clinical validity; the real cohort is tiny and single-centre; calibration not assessed; single-instance design; no browser end-to-end or penetration tests. For teaching and demonstration only, never for diagnosis or treatment.

Conclusion and Future Scope

An integrated, tested and honest engineering prototype. *Future work:* credentialed full MIMIC-IV with external validation and calibration, feature caching and worker processes, shared state for scaling, real document uploads, end-to-end and security testing.